

Spatial Forecast Errors and Renewable Curtailment

in ERCOT: A Functional Regression Analysis

April 2025 · Preliminary Results

Abstract. We estimate scalar-on-function regression models mapping 2D spatial weather forecast error fields (wind speed and temperature, HRRR 1h and GFS day-ahead) to system-wide renewable curtailment in ERCOT over six months of 2025 (4,364 hours, 6,502 infrastructure pixels). Partial Least Squares (PLS) explains 55% of curtailment variance from wind speed errors alone, outperforming FPCA (44%) by discovering that 40% of the curtailment-predictive signal lives in low-variance spatial modes that FPCA under-weights. Regime stratification reveals that temperature errors rival wind errors in extreme cold events ($R^2 \approx 0.57$ each), while day-ahead GFS errors dominate in extreme heat. A two-field MLP neural operator (wind + temperature, $4 \times$ downsampled grid) achieves CV $R^2 = 0.90$, establishing the strong nonlinearity of the spatial error–curtailment mapping.

1. Data and Setup

Sample. Six months of 2025 (January, March, May, July, September, November), yielding $T = 4,364$ hourly observations. Weather forecast errors are computed against ERA5-Land reanalysis on the 0.1° grid; infrastructure pixels are the $N = 6,502$ ERA5 cells containing at least one generating unit, transmission line, or load bus in ERCOT.

Forecast error fields. Four fields are analyzed:

- `wspd_error_1h`: HRRR 1h-ahead wind speed error (forecast – ERA5) [m/s]
- `temp_error_1h`: HRRR 1h-ahead temperature error [$^\circ\text{C}$]
- `wspd_error_0h`: GFS day-ahead wind speed error [m/s]
- `temp_error_0h`: GFS day-ahead temperature error [$^\circ\text{C}$]

Outcome. `total_curtailment_mw`: system-wide wind + solar curtailment (mean = 1,250 MW, std = 2,269 MW, min = 15 MW, max = 17,124 MW).

Evaluation. 5-fold cross-validated R^2 throughout, except regime analyses (small N) use 5-fold with $n=10$ PLS components. All models fit on standardized predictors.

2. Baseline Models: Ridge and FPCA

2.1. Ridge Regression (Raw Pixels)

A Ridge regression on all 6,502 pixel errors achieves $CV R^2 = 0.604 \pm 0.028$ — an upper bound for linear single-field methods, though numerically ill-conditioned (condition number $\approx 10^8$) and uninterpretable due to the high dimensionality.

2.2. Functional PCA (FPCA)

FPCA projects the $T \times N$ error matrix onto the top-K eigenvectors of the spatial covariance $X^T X$ and regresses curtailment on the resulting scores. The coefficient surface $\beta(s)$ is reconstructed as a linear combination of the K spatial modes.

K	CV R^2	$\pm$ Std
5	0.258	0.016
10	0.318	0.020
20	0.350	0.020
50	0.379	0.019
100	0.438	0.026
200	0.471	0.024

Table 1: FPCA performance vs number of components, `wspd_error_1h`. K=200 is the best interpretable model.

The $\beta(s)$ surface concentrates positive coefficients in the West Texas / Panhandle wind belt — regions where overpredicted wind speeds generate excess curtailment. High-impact pixels are stable across CV folds (651 pixels, 10% of the grid, identified consistently in all 5 folds).

3. Partial Least Squares: Supervised Basis Construction

PLS constructs basis vectors that maximize $\text{Cov}(Xw, Y)$ rather than $\text{Var}(X)$, directly targeting curtailment-predictive spatial patterns.

n components	CV R^2	$\pm$ Std	Coef instability
5	0.464	0.025	0.533
10	0.532	0.029	1.935
20	0.554	0.042	5.915
50	0.400	0.064	19.32
100	0.092	0.061	36.39

Table 2: PLS performance vs n components. n=20 is optimal; n $\geq$ 50 overfits severely.

PLS $n=20$ ($R^2=0.554$) outperforms FPCA $K=200$ ($R^2=0.471$) using only 10% as many features. The supervised basis construction is the single largest methodological improvement.

3.1. Why PLS and FPCA Find Different Patterns

The PLS and FPCA $\beta(s)$ surfaces have Pearson $r = 0.14$ — nearly orthogonal despite both predicting curtailment from the same data. Decomposing the FPCA score covariance with Y reveals:

- The top-20 variance components (PC 1–20) capture **74%** of total $|\text{Cov}(\vartheta_k, Y)|$ — but not all.
- **40% of the curtailment-predictive signal lives in low-variance FPCA components** (PC 22, 25, 30, 42, 45, 52, 54, 80+).
- Only 12/20 top-variance components overlap with the top-20 covariance-ranked components.

These low-variance but high-covariance modes likely represent fine-scale regional contrast patterns — opposing error signs across the wind belt versus load centers — that have weak grid-wide variance but strong predictive power because they drive local supply-demand imbalances. FPCA under-weights them; PLS explicitly seeks them.

4. All Four Error Fields

Field	FPCA R^2	$\pm$	PLS R^2	$\pm$	β corr
wspd_error_1h	0.438	0.025	0.554	0.042	0.142
wspd_error_0h	0.341	0.024	0.442	0.054	0.188
temp_error_1h	0.321	0.020	0.534	0.021	0.177
temp_error_0h	0.234	0.033	0.354	0.058	0.224

Table 3: FPCA ($K=100$) and PLS ($n=20$) performance across all four error fields. “ β corr” is the Pearson correlation between PLS and FPCA coefficient surfaces.

Key observations:

- **PLS outperforms FPCA by 10–21 pp** across all fields — the supervised advantage is universal.
- **Temperature errors are highly predictive for curtailment.** HRRR 1h temperature errors ($R^2=0.534$) are nearly as predictive as wind errors ($R^2=0.554$), which is surprising given that curtailment is directly driven by wind generation capacity.
- **Day-ahead GFS errors are weaker than 1h HRRR errors** across both wind and temperature, consistent with greater spatial specificity at shorter lead times.
- **β correlations are uniformly low** ($r=0.14$ – 0.22) — the divergence between supervised and unsupervised spatial patterns is systematic across all fields.

5. Extreme Weather Regime Stratification

Regime thresholds are computed from the full 4,409-hour dataset:

- **Extreme cold:** system-mean temperature $< 2.3^{\circ}\text{C}$ (5th percentile) — 212 hours (4.8%)
- **Extreme heat:** system-mean temperature $> 32.1^{\circ}\text{C}$ (95th percentile) — 212 hours (4.8%)

5.1. Extreme Cold

Field	FPCA R^2	$\pm$	PLS R^2	$\pm$
wspd_error_1h	0.553	0.117	0.629	0.144
temp_error_1h	0.567	0.103	0.606	0.106
wspd_error_0h	0.400	0.089	0.444	0.128
temp_error_0h	0.506	0.062	0.555	0.067

Table 4: Regime-stratified results for extreme cold hours (212 hours). FPCA $K=50$, PLS $n=10$.

Temperature errors rival wind errors in extreme cold. FPCA R^2 for temp_error_1h (0.567) slightly exceeds wspd_error_1h (0.553) — a complete reversal of the full-sample ordering where wind dominates. During cold snaps, demand spikes likely force dispatch decisions that interact strongly with renewable dispatch, making temperature forecast accuracy as operationally important as wind forecast accuracy.

The $\beta(s)$ surface for extreme cold has Pearson $r = 0.13$ with the full-sample surface — **near-orthogonal spatial attribution**. Full-sample models cannot recover the spatial structure relevant to cold-weather grid stress.

5.2. Extreme Heat

Field	FPCA R^2	$\pm$	PLS R^2	$\pm$
wspd_error_0h	0.515	0.186	0.567	0.156
wspd_error_1h	0.395	0.178	0.513	0.199
temp_error_1h	0.362	0.216	0.375	0.392
temp_error_0h	0.303	0.280	0.375	0.493

Table 5: Regime-stratified results for extreme heat hours (212 hours). Rows ordered by FPCA R^2 .

In extreme heat, **GFS day-ahead wind errors ($R^2=0.515$) outperform HRRR 1h errors ($R^2=0.395$)**. This reversal — shorter lead time being less predictive — is consistent with day-ahead unit commitment driving outcomes during summer heat events: the day-ahead schedule is set before the heat peak emerges, so day-ahead forecast accuracy for wind (which affects how much thermal capacity is committed) matters more than real-time accuracy. Temperature errors are substantially weaker in extreme heat ($R^2 \approx 0.30\text{--}0.37$), consistent with the mechanism being wind-driven curtailment rather than demand-driven dispatch.

The extreme heat $\beta(s)$ surface has $r = 0.42$ with the full-sample surface — more similar than extreme cold, but still substantially different.

5.3. Quantile Regression on Curtailment Extremes

Using FPCA scores ($K=100$) as predictors for quantile regression on `wspd_error_1h`:

Quantile τ	Pseudo- R^2
0.25	0.206
0.50	0.183
0.75	0.387
0.90	0.651

Table 6: Quantile regression pseudo- R^2 by target quantile.

Extreme curtailment events ($\tau=0.90$) are $3.5\times$ more predictable from spatial forecast errors than median curtailment. This is the most operationally significant finding: the grid events that matter most (high curtailment) are precisely the ones where spatial forecast error structure is most predictive.

6. Neural Operators: Nonlinear Mapping

To test whether the spatial error–curtailment relationship is linear, we train three neural architectures on a 2-channel input grid (wind + temperature error, HRRR 1h) downsampled to $27\times 33 = 891$ cells ($4\times$ spatial stride).

Architecture	CV R^2	$\pm$ Std	Notes
MLP	0.901	0.013	$1782 \rightarrow 256 \rightarrow 256 \rightarrow 64 \rightarrow 1$
FNO	0.736	0.038	2-layer, modes=8, hidden=32
CNN	0.488	0.059	3-layer conv + global avg pool
Ridge (baseline)	0.604	0.028	single-field, 6,502 pixels
PLS $n=20$ (baseline)	0.554	0.042	single-field, wspd only

Table 7: Neural operator results vs linear baselines. 3-fold CV, 50 epochs, early stopping. MLP uses flattened $27\times 33\times 2 = 1,782$ inputs; FNO uses spectral convolutions; CNN uses local 3×3 convolutions.

6.1. Key Findings

MLP ($R^2=0.901$) reduces unexplained variance by 50% relative to the best linear model (Ridge $R^2=0.604$). Combining wind and temperature error spatial fields with a nonlinear architecture captures interaction effects that no linear $\beta(s)$ model can represent.

FNO ($R^2=0.736$) outperforms all linear baselines despite operating on $16\times$ downsampled data. Fourier spectral convolutions capture multi-scale spatial structure that local pixel regres-

sion misses — consistent with curtailment being driven by large-scale error patterns spanning the entire West Texas wind belt.

CNN ($R^2=0.488$) underperforms both. Local 3×3 receptive fields are insufficient; global spatial structure — either via full-grid flattening (MLP) or spectral decomposition (FNO) — is the key inductive bias for this problem.

The multi-field nonlinear advantage is large. Moving from single-field PLS ($R^2=0.554$) to two-field MLP ($R^2=0.901$) represents the combined effect of (a) adding temperature errors, and (b) allowing nonlinear interactions. Runs 1–3 showed temperature adds 0.05–0.10 R^2 in linear models; the bulk of the remaining gain (0.30 R^2) must come from nonlinearity.

Caveat. All neural results use $4\times$ spatial downsampling and CPU training (50 epochs, 3-fold CV). Full-resolution GPU training would yield higher baselines; results here are conservative lower bounds on neural operator performance.

7. Spatial Smoothness Regularization

We attempted to impose spatial smoothness on PLS loading vectors via a Laplacian penalty: projected weight vectors through $(I + \lambda L)^{-1}$ at each NIPALS iteration, where $L = D - A$ is the spatial graph Laplacian on the ERA5 pixel neighborhood graph ($6,502 \times 6,502$, 57,114 nonzeros).

λ	n=5	n=10	n=20	n=50
0.00	0.464	0.532	0.554	0.400
0.01	0.128	0.104	−0.241	−4.142
0.10	0.120	0.110	−0.120	−2.729
1.00	0.092	0.117	0.046	−0.712
10.00	0.038	0.089	0.098	−0.113

Table 8: Constrained PLS CV R^2 by λ and n components. Standard PLS ($\lambda=0$) is the only stable configuration.

Penalized NIPALS is numerically unstable for any $\lambda > 0$. The loading matrix reconstruction $W(P^\top W)^{-1}Q$ becomes ill-conditioned when smoothed weight vectors are near-collinear, causing R^2 to collapse to as low as −4.14. Post-hoc smoothing of the $\beta(s)$ surface (graph total variation penalty after fitting standard PLS) is the numerically robust alternative for interpretability.

8. Summary and Implications

Model / Setting	CV R^2	Key feature
MLP (wind + temp, 2-field)	0.901	nonlinear, multi-field
FNO (wind + temp, downsampled)	0.736	spectral, multi-field
Ridge (wind, raw pixels)	0.604	linear, full-res, single-field
PLS $n=20$ (wind, HRRR 1h)	0.554	linear, supervised basis
FPCA $K=200$ (wind)	0.471	linear, unsupervised basis
PLS, extreme cold	0.629	regime-specific
PLS, extreme heat	0.513	regime-specific
Quantile $\tau=0.90$ (FPCA)	0.651	tail events

Table 9: Summary of best-performing models across all analyses.

Five main conclusions:

- Spatial forecast errors are strongly predictive of curtailment ($R^2=0.55\text{--}0.90$).**
The ERA5 0.1° error field carries substantial operational signal well beyond what aggregate error statistics would suggest.
- PLS decisively outperforms FPCA in the linear regime** because 40% of the curtailment-predictive signal lies in low-variance spatial modes that FPCA under-weights. Supervised basis construction is the most impactful methodological choice in the linear setting.
- Temperature errors are nearly as important as wind errors**, especially in extreme cold ($R^2=0.57$ for temp vs 0.55 for wind). Operational models that focus only on wind forecast errors miss a significant fraction of predictable curtailment.
- Regime-specific spatial attribution is essential.** Full-sample $\beta(s)$ surfaces share almost no structure with extreme-cold $\beta(s)$ surfaces ($r=0.13$). Grid operations during cold snaps are driven by a fundamentally different spatial error pattern than normal conditions.
- The error–curtailment mapping is strongly nonlinear.** A simple two-field MLP achieves $R^2=0.90$, reducing unexplained variance by 50% relative to linear Ridge. The next research priority is understanding which interactions drive the nonlinear gain — likely threshold effects in dispatch constraints and transmission binding — and whether regime-stratified nonlinear models yield further improvements.

9. Appendix: Model Specifications

FPCA. Truncated SVD on the standardized $T \times N$ error matrix, retaining top- K left singular vectors as spatial modes. Ridge regression on K scores.

PLS. NIPALS algorithm. Loading vectors w maximize $\text{Cov}(Xw, Y)$ subject to $\|w\|=1$ and orthogonality to prior components. Scalar regression on projected scores.

Ridge. 5-fold CV over $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ on raw standardized pixel values.

MLP. Architecture: $\text{flatten}(C \times H \times W) \rightarrow \text{Linear}(1782, 256) \rightarrow \text{GELU} \rightarrow \text{Linear}(256, 256) \rightarrow \text{GELU} \rightarrow \text{Linear}(256, 64) \rightarrow \text{GELU} \rightarrow \text{Linear}(64, 1)$. Adam lr=1e-3, cosine LR decay, early stopping patience=10.

FNO. $\text{Lift} \rightarrow 2 \times [\text{SpectralConv2d}(\text{modes}=8) + \text{Conv2d}(1 \times 1) + \text{GELU}] \rightarrow \text{global average pool} \rightarrow \text{MLP}(32 \rightarrow 16 \rightarrow 1)$. SpectralConv2d operates on the top-8 Fourier modes in each spatial dimension.

CNN. $\text{Conv2d}(2 \rightarrow 32, 3 \times 3) \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32 \rightarrow 32, 3 \times 3) \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32 \rightarrow 16, 3 \times 3) \rightarrow \text{GELU} \rightarrow \text{global avg pool} \rightarrow \text{Linear}(16 \rightarrow 32) \rightarrow \text{GELU} \rightarrow \text{Linear}(32 \rightarrow 1)$.

Data. 6 months (Jan, Mar, May, Jul, Sep, Nov 2025). ERA5 0.1° grid, N=6,502 infrastructure pixels. Neural operator uses 4× spatial downsampling ($105 \times 130 \rightarrow 27 \times 33$). All Y normalized to zero mean / unit variance for neural training; R^2 computed on original scale.